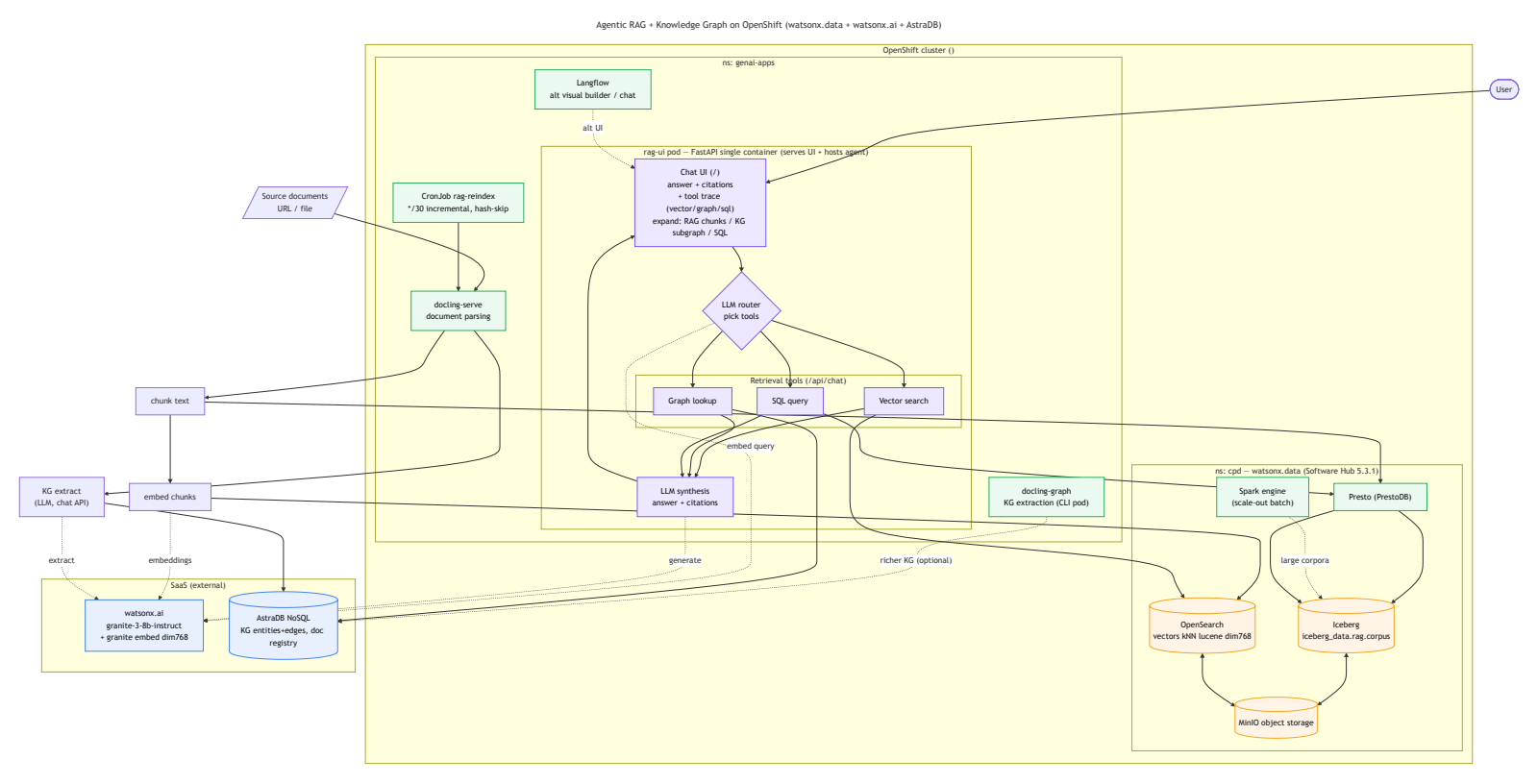
Agentic RAG + Knowledge Graph on OpenShift (watsonx.data + watsonx.ai + AstraDB)
OpenShift cluster ()
ns: genai-apps
Langflow
alt visual builder / chat
alt UI
User
rag-ui pod — FastAPI single container (serves UI + hosts agent)
Chat UI (/)
answer + citations
+ tool trace
(vector/graph/sql)
expand: RAG chunks / KG
subgraph / SQL
Source documents
URL / file
CronJob rag-reindex
*/30 incremental, hash-skip
docling-serve
document parsing
LLM router
pick tools
Retrieval tools (/api/chat)
Graph lookup
SQL query
Vector search
chunk text
embed query
KG extract
(LLM, chat API)
embed chunks
LLM synthesis
answer + citations
docling-graph
KG extraction (CLI pod)
ns: cpd — watsonx.data (Software Hub 5.3.1)
Spark engine
(scale-out batch)
Presto (PrestoDB)
extract
embeddings
generate
richer KG (optional)
large corpora
SaaS (external)
watsonx.ai
granite-3-8b-instruct
+ granite embed dim768
AstraDB NoSQL
KG entities+edges, doc
registry
OpenSearch
vectors kNN lucene dim768
Iceberg
iceberg_data.rag.corpus
MinIO object storage